Clinical Model Performance Report

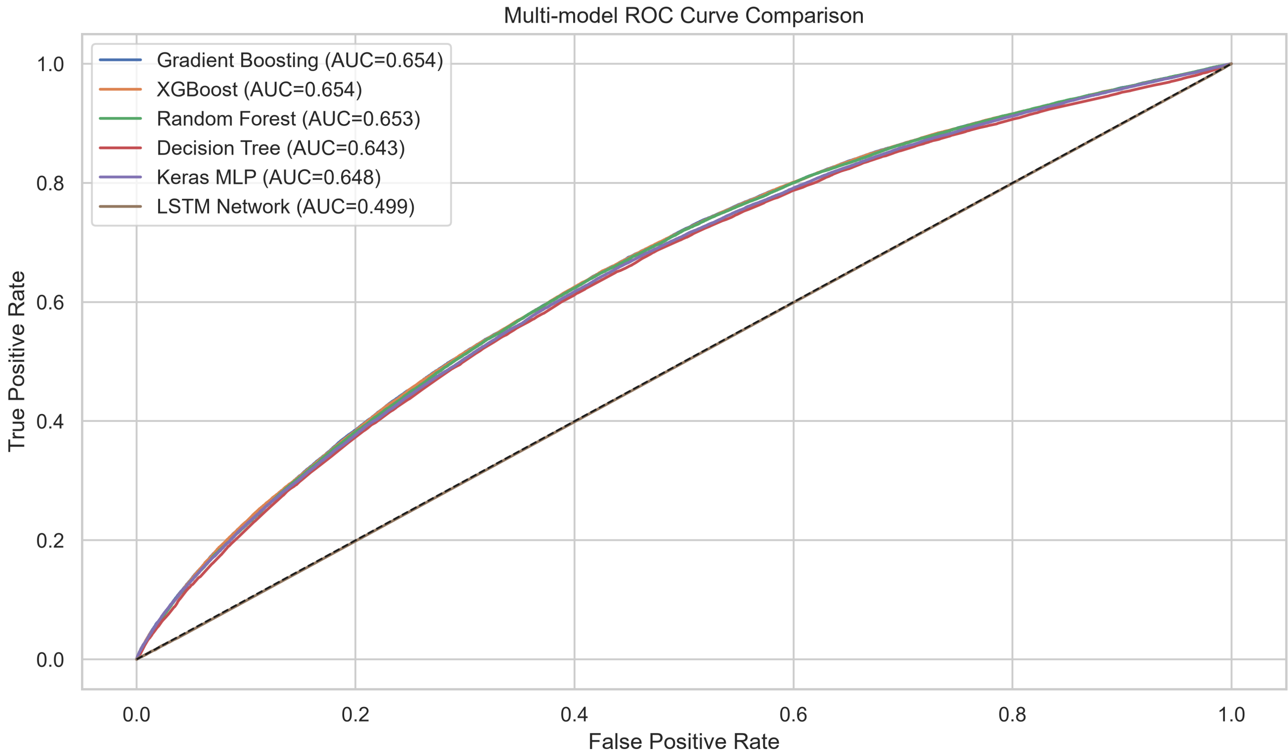
Executive summary: Gradient Boosting ranked first by PR-AUC, then F1-Score and Recall. The selected threshold was tuned only on an internal validation partition; the final test set remained untouched until evaluation.

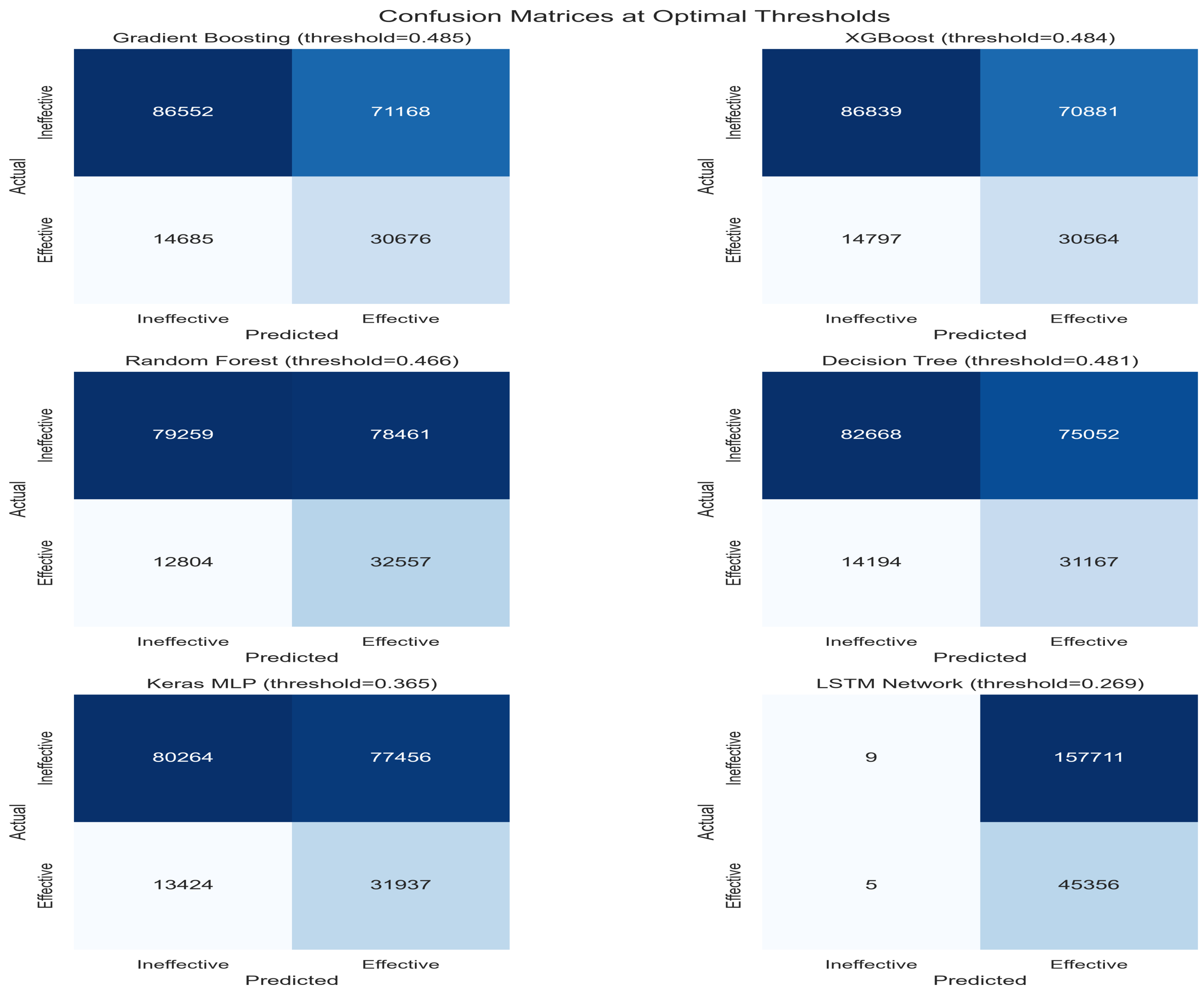
Performance Benchmark

Rank	Model	PR-AUC	F1-Score	Recall	ROC-AUC	Accuracy	Precision	Threshold
1	Gradient Boosting	0.3459	0.4168	0.6763	0.6543	0.5772	0.3012	0.4849
2	XGBoost	0.3453	0.4164	0.6738	0.6541	0.5781	0.3013	0.4844
3	Random Forest	0.3426	0.4164	0.7177	0.6526	0.5506	0.2933	0.4663
4	Keras MLP	0.3411	0.4127	0.7041	0.6483	0.5525	0.2919	0.3646
5	Decision Tree	0.3302	0.4112	0.6871	0.6431	0.5605	0.2934	0.4806
6	LSTM Network	0.2231	0.3651	0.9999	0.4989	0.2234	0.2234	0.2693

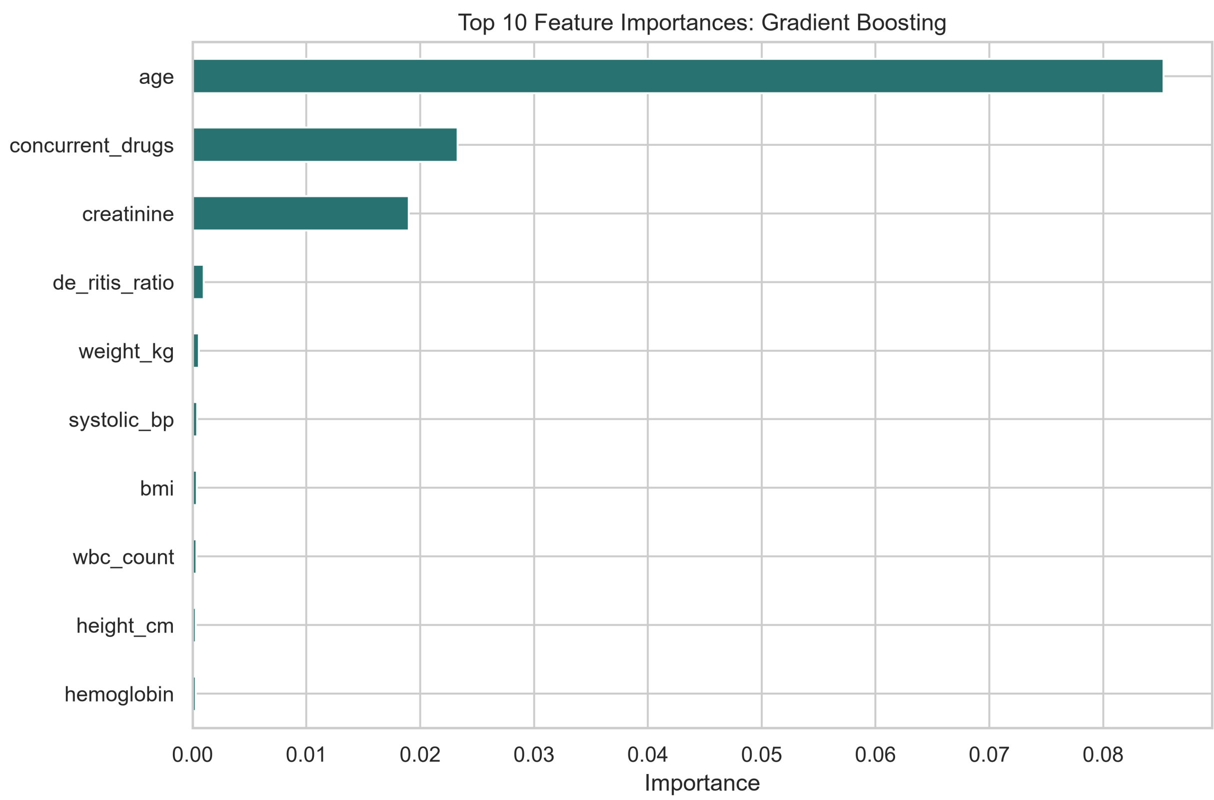
Overfitting Analysis

Model	Train ROC-AUC	Test ROC-AUC	Gap	Status
Gradient Boosting	0.6652	0.6543	0.0109	Well-Balanced
XGBoost	0.6735	0.6541	0.0194	Well-Balanced
Random Forest	0.7007	0.6526	0.0481	Moderate Overfit
Keras MLP	0.6510	0.6483	0.0028	Well-Balanced
Decision Tree	0.6705	0.6431	0.0275	Well-Balanced
LSTM Network	0.4993	0.4989	0.0004	Well-Balanced





Confusion Matrices at Optimal Thresholds
Gradient Boosting (threshold=0.485)
86552
71168
14685
30676
XGBoost (threshold=0.484)
86839
70881
14797
30564
Random Forest (threshold=0.466)
79259
78461
12804
32557
Decision Tree (threshold=0.481)
82668
75052
14194
31167
Keras MLP (threshold=0.365)
80264
77456
13424
31937
LSTM Network (threshold=0.269)
9
157711
5
45356
Actual
Ineffective
Effective
Predicted



Top 10 Feature Importances: Gradient Boosting
age
concurrent_drugs
creatinine
de_ritis_ratio
weight_kg
systolic_bp
bmi
wbc_count
height_cm
hemoglobin
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Importance